

Dynamic NPC AI System (Text-to-Text)

System Purpose

The Dynamic NPC AI system is designed as a **system showcase**, where AI-driven interaction itself functions as the core mechanic rather than a supporting feature. The goal of this system is to demonstrate how a shared AI framework can generate **distinct conversational behavior** across multiple NPCs using attribute-based logic instead of traditional scripted dialogue trees.

This system primarily exists to validate:

- Attribute-driven response variation
- Scalable NPC architecture
- Immersion-consistent AI interaction
- Clear technical separation between core logic and interface layers

At its current stage, the system is intentionally scoped as a **technical and experiential prototype**, not a complete gameplay loop.

NPC Architecture Overview

- **Total NPCs:** 4
- **System Type:** Shared AI system with per-NPC attribute configuration
- **NPC Type:** Human
- **Setting:** Medieval-inspired environment

All NPCs operate on the same underlying AI logic. Behavioral variation is achieved exclusively through unique attribute assignments, allowing the system to scale without duplicating logic or content.



NPC World Behavior

NPCs exist as active entities within the environment rather than static interaction points.

- NPCs roam freely within their designated world space
- Roaming behavior is autonomous and continuous
- When the player approaches an NPC:
 - NPC movement is paused
 - A dialogue interface is triggered
 - Text-to-text communication begins

This approach prioritizes **world believability** and avoids menu-driven interaction patterns.

Attribute-Based Response System

Each NPC is defined by a unique set of attributes that influence how responses are generated. These attributes do not change during gameplay in the current implementation.

Example Attribute Categories

NPC_Name	Eldrin	🚩
NPC_Role	Wise mage and advisor	🚩
NPC_Personality	Loyal, poetic, and patient	🚩
NPC_SpeechStyle	Formal and ancient	🚩
NPC_CurrentMood	Calm but slightly worried	🚩
NPC_ThemeSetting	Dark fantasy world filled with magic	🚩
NPC_Goal	Guide the player through the curse	🚩
NPC_Backstory	Once a royal court mage, now a rec	🚩

The same player input produces different responses depending on the NPC's attribute configuration. This demonstrates behavioral diversity without branching dialogue trees or hand-authored responses.

Player Interaction Model

- **Input Type:** Free-form text
- **Output Type:** AI-generated text
- **Interaction Trigger:** Player proximity to NPC
- **UI Style:** Medieval-themed dialogue interface

There are no predefined dialogue choices. The player may ask any question, enabling open-ended experimentation with the AI system.

Memory & Persistence (Current Implementation)

- NPCs do not retain memory of past interactions
- No short-term or long-term conversational memory
- Each interaction is stateless

This limitation is intentional for the current prototype phase, allowing focused evaluation of attribute-driven behavior without compounding system complexity.

Consequences & Game State Impact (Current Implementation)

- No world-state changes
- No progression gates or unlocks
- No failure or success conditions
- Dialogue does not alter NPC behavior after interaction ends

The system currently functions as a **sandbox AI interaction layer** rather than a full gameplay mechanic.

Tone & Immersion Alignment

- NPC communication tone is derived entirely from assigned attributes
- Language style and phrasing are tuned to match the medieval setting
- UI presentation reinforces environmental immersion

The system prioritizes thematic consistency over mechanical depth at this stage.

Technical Characteristics

- **AI Type:** Online (LLM-based)
 - **Response Model:** Probabilistic, attribute-conditioned
 - **System Role:** Core technical showcase
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Future Scope (Planned – Not Implemented)

This section documents planned extensions for future development and serves as a reference for long-term system evolution. None of the following features are part of the current implementation.

Voice-to-Voice Interaction

- Speech-to-Text (STT) replaces player text input
- Text-to-Speech (TTS) replaces NPC text output
- Core AI logic remains unchanged

Conversational Memory

- NPCs retain memory of:
 - Previous questions
 - Player tone and intent
 - Logical contradictions
- Memory influences future responses and trust levels

Meta AI & Visual Feedback

- Facial animation and lip-sync driven by dialogue output
 - Emotion-aware delivery through tone modulation
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Planned Gameplay Concept Using This System

Core Concept

- **Player Role:** Investigator
- **NPC Count:** 4
- **Environment:** Single interrogation space
 - Table and two chairs
 - Blacked-out surroundings
 - Headlamp as the only light source

Objective

Identify the criminal among the four NPCs using **conversation alone**.

- No traditional UI overlays

- No physical interaction mechanics
- No explicit prompts or markers

Design Intent

- Progression is driven entirely by questioning and inference
 - NPCs may lie, evade, or manipulate based on personality and memory
 - Truth is never directly revealed and must be deduced by the player
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Current Evaluation

At its present stage, the Dynamic NPC AI system represents a **completed system prototype**, not a full game. Its strength lies in architectural clarity, scalability, and a clearly defined roadmap for expansion rather than immediate gameplay depth.

This document intentionally separates **current implementation** from **future ambition** to maintain technical honesty and long-term usability.